Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY **9648/02**

Paper 2 Core Paper (Booklet I) **16 Sep 2011**

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST.
Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A:
Consists of <u>8</u> Structured Questions separated into Booklets I and II.
Answer **all** questions.
Write your answers in the **space provided** on the question paper.

Section B:
Consists of <u>2</u> Essay Questions found in Booklet II.
Answer any **one** question.
Write your answers on the separate **writing papers** provided.
At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [ ] at the end of each question or part question.

Do not open the booklets until you are told to do so.

BOOKLET I		
FOR EXAMINER'S USE		
Section A1	1	/10
	2	/10
	3	/10
	4	/10

This document consists of **9** printed pages and **1** blank page.

Booklet I

Section A1: Structured Questions (40 marks)

Answer **all** questions in this section.

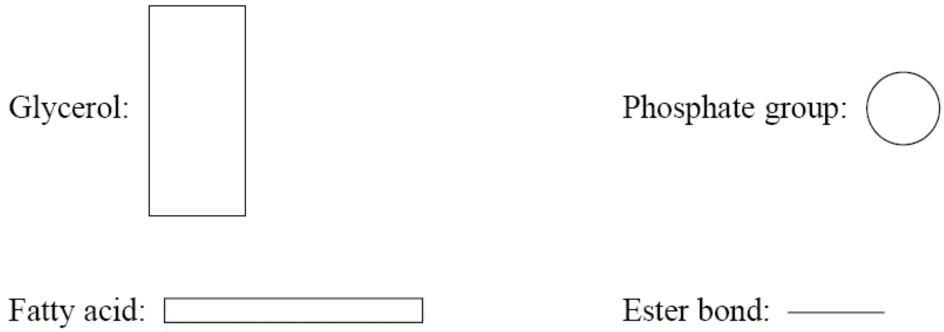
For Examiner's use

Question 1

(a) (i) Give **two** differences in the molecular structure of the two components of starch, **amylose** and **amylopectin**. [2]

(ii) Explain how the structure of starch relates to its function in plants. [2]

(b) **(i)** In the space provided below, draw a diagram to show the structure of a phospholipid. Use the symbols shown for each component in your diagram. [3]



(ii) The presence of phosphate in a phospholipid makes part of the molecule hydrophilic. Explain what is meant by the term ***hydrophilic***. [1]

(iii) Describe the role of phospholipids in the cell surface membrane. [2]

Total: [10]

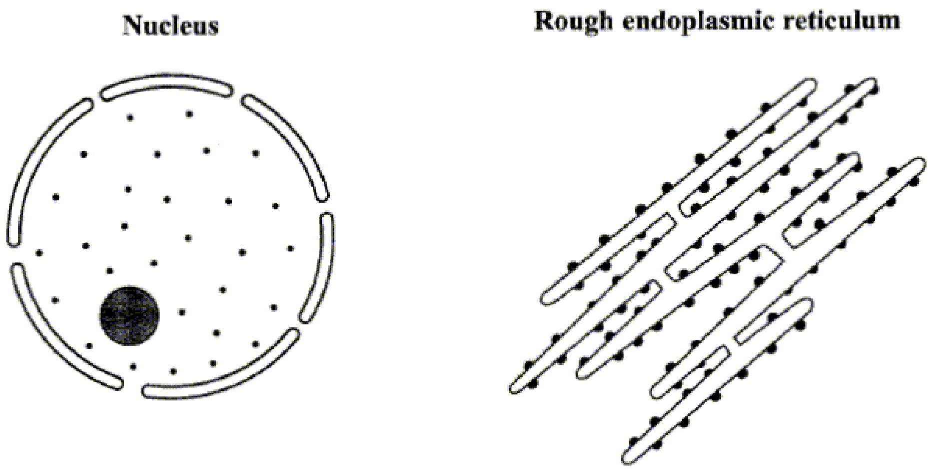
Question 2

For Examiner's use

(a) The table below refers to some cell structures. Complete the table below by inserting the correct word, words or diagram in the appropriate boxes. Leave the shaded boxes empty. [4]

Name of cell structure	Description of cell structure	Drawing of cell structure
(i)	1. Darkly stained region in the nucleus. 2. Where ribosome RNA is made.	
Centriole		(ii)
Lysosome	(iii) 1. 2.	
(iv)	1. Hollow cylinders made of protein. 2. Form spindle fibres.	

(b) **(i)** The diagram below shows some cell structures involved in protein synthesis in eukaryotic cells.



Describe the events that occur inside the nucleus to produce a molecule of messenger RNA (mRNA). [4]

(ii) Describe the role of the ribosome in protein synthesis. [2]

Total: [10]

Question 3

For Examiner's use

In maize, the seeds can be yellow or white in colour. In addition, the seeds may have a smooth surface or a wrinkled surface.

Each of these characteristics of maize seeds is an example of single-gene inheritance.

If a pure-breeding (homozygous) variety of maize with yellow, smooth seeds is crossed with a pure-breeding variety with white, wrinkled seeds, all of the F_1 generation have yellow, smooth seeds.

(a) **(i)** Suggest suitable symbols that could be used in a genetic diagram for the alleles involved in these characteristics. [1]

Allele for yellow colour : ____________________

Allele for white colour : ____________________

Allele for smooth seed : ____________________

Allele for wrinkled seed : ____________________

(ii) Using these symbols, give the genotype of both the pure-breeding varieties. [2]

Variety producing yellow, smooth seeds : ____________________

Variety producing white, wrinkled seeds : ____________________

(b) **(i)** A test cross was carried out on the F_1 seeds. The numbers of seeds resulting from this cross are shown in the table below.

The χ^2 test can be used to test the hypothesis that there is no significant difference between these results and the expected 1 : 1 : 1 : 1 ratio.

Complete the table below to calculate the value of χ^2 for these results. [2]

Phenotype	**Observed (O)**	**Expected (E)**	**(O – E)**	**$(O - E)^2$**	**$\frac{(O - E)^2}{E}$**
Yellow, smooth	2046				
White, smooth	851				
Yellow, wrinkled	750				
White, wrinkled	1953				
Total number of seeds	5600	5600	$\Sigma \frac{(O - E)^2}{E} =$		

(ii) The table shows values for χ^2 at different levels of probability and for different degrees of freedom.

Degrees of freedom	**Probability, *p***				
	0.2	**0.1**	**0.05**	**0.02**	**0.01**
1	1.64	2.71	3.84	5.41	6.64
2	3.22	4.61	5.99	7.82	9.21
3	4.64	6.25	7.82	9.84	11.35
4	5.99	7.78	9.49	11.67	13.28
5	7.29	9.24	11.07	13.39	15.09

What should the breeders conclude about the significance of their results? Explain your answer. [2]

(iii) With reference to the events that occur during meiosis, suggest why the results for the observed frequencies differ considerably from those for the expected frequencies. [3]

Total: [10]

For Examiner's use

Question 4

The bone protein, osteocalcin, has been extracted from fossil Neanderthal skulls found in Shanidar Cave, Iraq. The skulls were estimated to be approximately 75 000 years old.

Using mass spectrometry, the amino acids of this Neanderthal osteocalcin have been sequenced. This sequence has been compared with that in osteocalcin from *Homo sapiens* and other modern primates.

The sequences of the first 20 amino acids in the primary structure of osteocalcin from this study are shown below. Amino acids are represented by capital letters.

	1									10										20
Modern human	Y	L	Y	Q	W	L	G	A	P	V	P	Y	P	D	P	L	E	P	R	R
Neanderthal	Y	L	Y	Q	W	L	G	A	P	V	P	Y	P	D	P	L	E	P	R	R
Chimpanzee	Y	L	Y	Q	W	L	G	A	P	V	P	Y	P	D	P	L	E	P	R	R
Orang-utan	Y	L	Y	Q	W	L	G	A	P	V	P	Y	P	D	P	L	E	P	K	R
Gorilla	Y	L	Y	Q	W	L	G	A	O	V	P	Y	P	D	P	L	E	P	K	R
Monkey	Y	L	Y	Q	W	L	G	A	O	A	P	Y	P	D	P	L	E	P	K	R

(a) Explain the significance of each of the following in the formation of the primary structure of osteocalcin.

(i) A gene [2]

____

____

____

(ii) Peptide bonding [1]

____

____

____

(b) Suggest what the data indicate about the relationships between Neanderthals and modern primates. Give reasons for your answer. [4]

(c) Explain how material from the Neanderthal skulls could be used to estimate their age. [3]

END OF BOOKLET I

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Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY 9648/02

Paper 2 Core Paper (Booklet II) **16 Sep 2011**

2 hours

Additional Materials: Writing paper

BOOKLET II		
FOR EXAMINER'S USE		
Section A2	**5**	**/10**
	6	**/10**
	7	**/11**
	8	**/9**
Section B	**9 / 10***	**/20**

* Please circle the appropriate number.

This document consists of **11** printed pages and **3** blank pages.

Booklet II

Section A2: Structured Questions (40 marks)

Answer **all** questions in this section.

For Examiner's use

Question 5

Fig. 5.1 below shows how the amount of DNA in a cell of *Chrysanthemum makinoi* varies during periods **P** to **T** in the cell cycle.

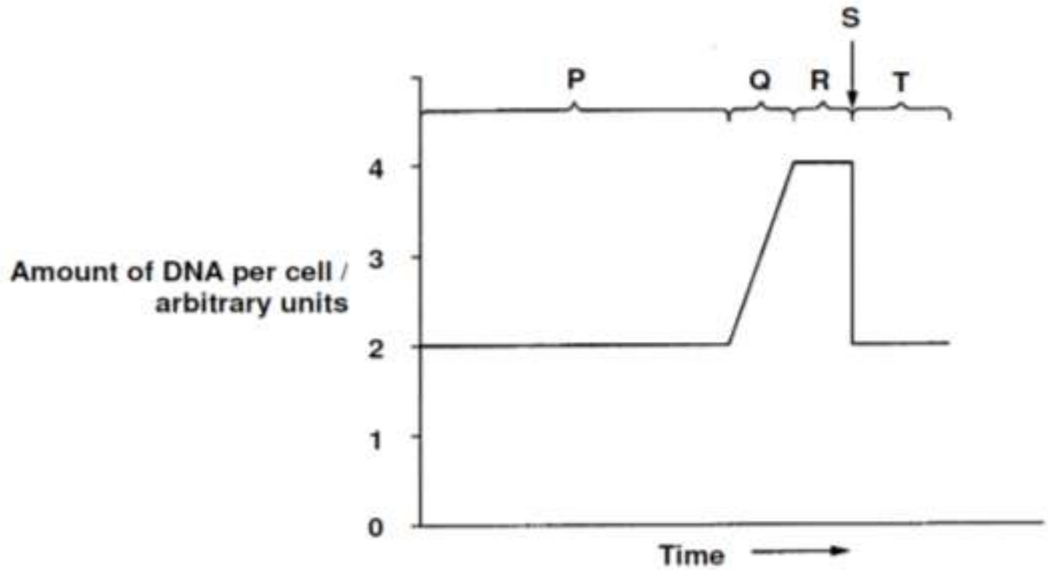

Fig. 5.1

(a) Radioactive thymine was supplied to the cells of some growing *Chrysanthemum* tissue.

With reference to **Fig. 5.1**, during which period (**P** to **T**) of the cell cycle does radioactivity of the nuclei first increase? Explain your answer. [3]

(b) At anaphase of mitosis, sister chromatids separate to become individual chromosomes.

(i) Distinguish between ***chromatids*** and ***chromosomes***. [2]

(ii) Identify the period (**P** to **T**) in **Fig. 5.1** during which anaphase occurs. Briefly explain your answer. [2]

(iii) With reference to **Fig. 5.1**, describe what would happen if a chromosome loses its centromere. [2]

(iv) Suggest why the loss of a centromere would theoretically not likely affect the ability of the cell to manufacture the necessary gene products. [1]

Total: [10]

For Examiner's use

Question 6

(a) **Fig. 6.1** shows a scanning electron micrograph of *Escherichia coli* under mass attack by numerous phage-T4 virions.

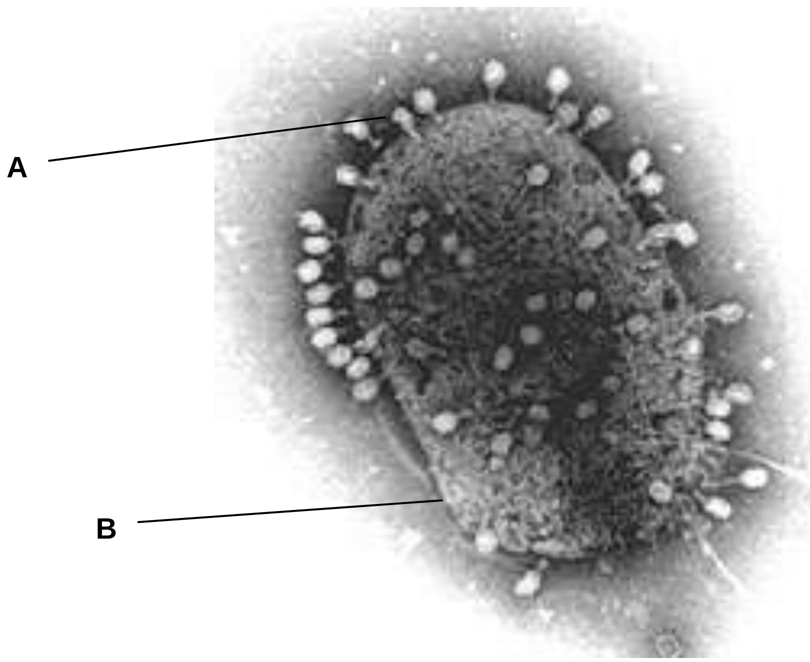

Fig. 6.1

(i) Suggest how **A** recognizes the correct host cell. [1]

____

____

(ii) With reference to **Fig. 6.1**, describe a process whereby bacterial DNA can be transferred from one cell to another. [3]

____

____

____

____

____

____

____

____

(iii) In the table below, give two differences between **two other processes** (besides the one named in **a (ii)**) that result in genetic variation in bacteria. [3]

Named Process 1:	Named Process 2 : ....
1 ________________________	________________________
2 ________________________	________________________

(b) While the processes in part **(a)** bring about genetic variation, the reproduction of bacteria does not lead to genetic variation.

(i) State the process whereby bacteria reproduce. [1]

____

(ii) Give **two** advantages of this form of reproduction to bacteria. [2]

____

____

____

____

Total: [10]

For Examiner's use

Question 7

Fig. 7.1 shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.

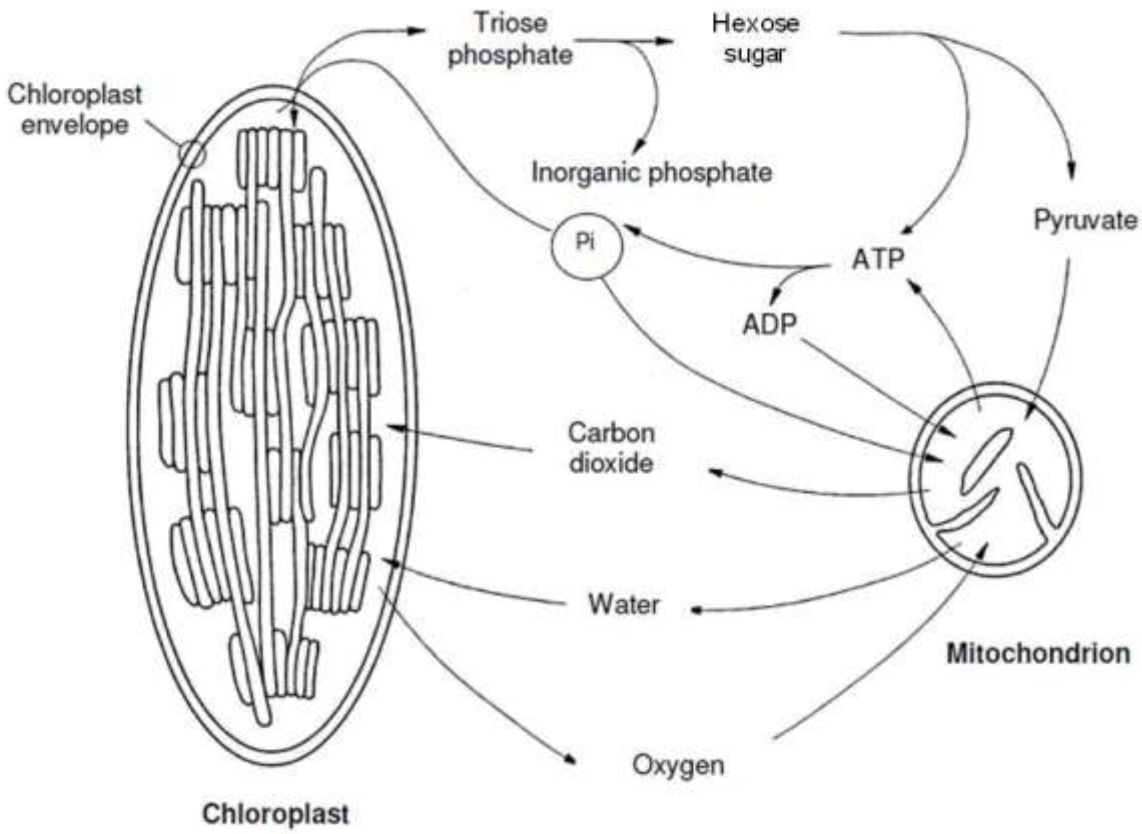

Fig. 7.1

A leafy shoot is sealed inside an air-tight transparent container. The concentration of oxygen in the atmosphere within this container can be measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Fig. 7.1**, outline how the following will link the metabolic pathways between the chloroplast and mitochondrion of this cell:

(i) triose phosphate and hexose sugar; [2]

(ii) gaseous exchange. [2]

(b) With reference to **Fig. 7.1**,

(i) explain how it is possible for the oxygen concentration to remain constant. [2]

(ii) explain why there is no build up in the concentration of phosphate ions inside the mitochondria as a result of the inward passage of phosphate ions. [2]

(c) An aqueous solution contains small membranous vesicles which may be derived either from a thylakoid or a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell.

Fig. 7.2 below shows the various experimental systems that are prepared in an attempt to synthesize ATP.

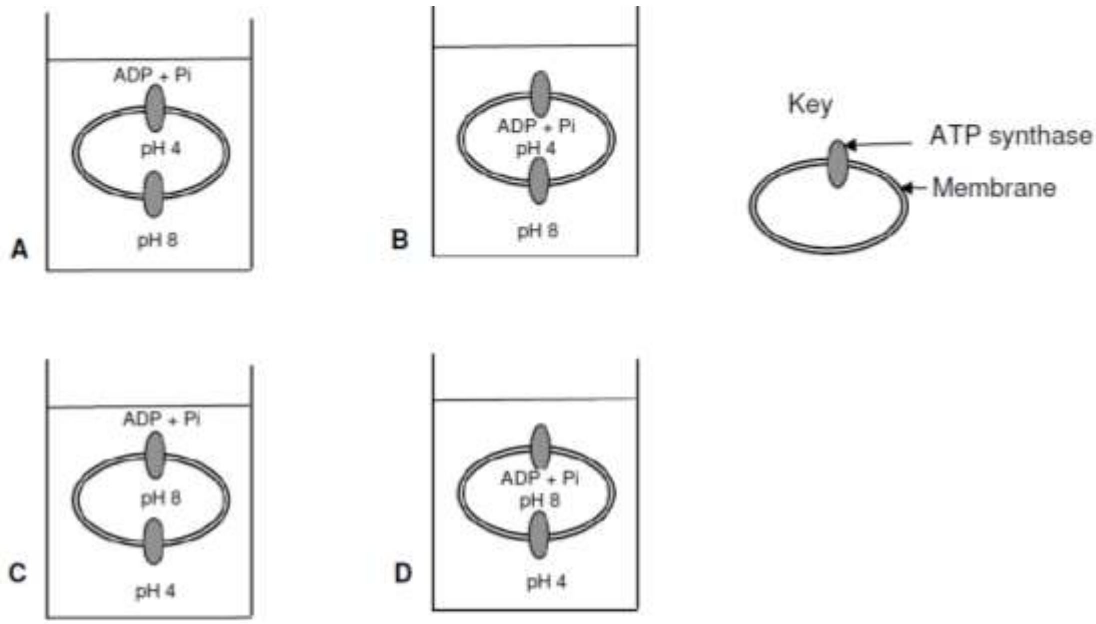

Fig. 7.2

Which experimental system (**A** to **D**) shown in **Fig. 7.2** would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer. [3]

Total: [11]

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For Examiner's use

Question 8

(a) Using an example from the generation of action potentials, explain what is meant by ***positive feedback***. [3]

(b) Myasthenia gravis is a disease which causes muscular weakness. It develops because of an attack by the body's own immune system on neuromuscular junctions. **Fig. 9.1** below shows a normal neuromuscular junction (nerve to muscle) and one affected by the disease (myasthenic).

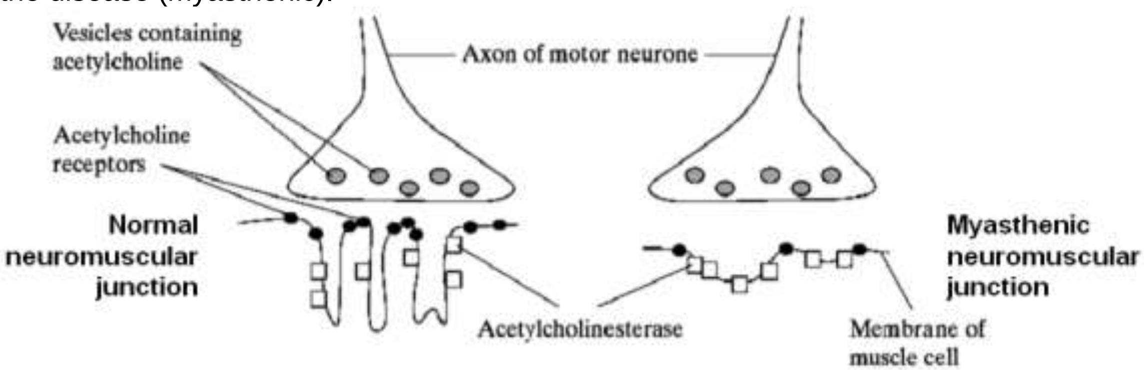

Fig. 9.1

Describe **two** ways in which a myasthenic neuromuscular junction differs from a normal one and explain how each difference would affect transmissions across the myasthenic neuromuscular junction. [4]

1.

2.

(c) Multiple sclerosis is a disease in which the myelin sheath surrounding neurones are destroyed and so the neurones become de-myelinated. People with multiple sclerosis produce slow responses to stimuli.

Explain how myelination of neurones helps in speeding up responses. [2]

Total: [9]

Section B: Essay Questions (20 marks)
Answer only **<u>one</u>** question.
Write your answers on the writing paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
A **NIL RETURN** is required.

Question 9

(a) Describe control elements and explain how they influence transcription. [7]

(b) Contrast between the prokaryotic and eukaryotic genomes. [7]

(c) Explain how changes at the genomic level can lead to cancer. [6]

<u>OR</u>

Question 10

(a) Describe, with named examples, how the **three** forms of natural selection bring about evolution. [10]

(b) Explain how species are formed with reference to physiological isolation. [10]

[Total: 20]

END OF PAPER